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SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Restaurant App MVP

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Course*: Mobile App Development (Fall 2026)*

Instructor*: MISS SADAF TANVIR*

1. Scope (Section 1.2)

- **In Scope (New Addition):** Express.js REST API Backend server and MongoDB database integration for persistent data sync across multiple customer and manager devices.
- **Out of Scope (Maintained):** Real third-party payment gateway integration (e.g., Stripe) and Push Notifications.
- **Problems Identified in A1 & Solutions:**
    1. *Issue:* Price changes by the manager do not sync to customer phones.

       Solution: Fetch menu items dynamically from MongoDB via /api/menu on app load/refresh.
    2. *Issue:* App reinstallation deletes order history and table reservations.

       Solution: Store orders and reservations permanently in MongoDB linked to user IDs.
    3. *Issue:* Orders placed on one device are not visible on other logged-in devices.

       Solution: Implement server-side persistence with polling on the Order Tracking screen.

2. Definitions (Section 1.3)

1. **API (Application Programming Interface):** A software intermediary allowing the React Native app to communicate with the Express server.

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2. **REST (Representational State Transfer):** An architectural style using standard HTTP methods (GET, POST, PUT, PATCH, DELETE).
3. **Endpoint:** A specific URI route on the server where API requests are sent (e.g., /api/menu).
4. **JSON (JavaScript Object Notation):** The lightweight data format used for server-client communication.
5. **MongoDB & Collection:** A NoSQL document database; collections represent logical groups like users, menuitems, orders.
6. **Mongoose:** An Object Data Modeling (ODM) library for MongoDB and Node.js.
7. **JWT (JSON Web Token):** A compact token mechanism used for stateless, secure user authentication.

3. Database Design (Replaces Section 5)

Collection	Fields & Types	Relationships & Rules
Users	_id (ObjectId), name (String, required), email (String, required, unique), password (String, required, hashed), role (String, enum: ['customer', 'manager'], default: 'customer'), createdAt (Date)	Referenced by Orders and Reservations.
MenuItems	_id (ObjectId), name (String, required), description (String), price (Number, required, min: 0), category (String, required), image (String), isSpecial (Boolean, default: false), isAvailable (Boolean, default: true)	Referenced inside Order item snapshots.
Tables	_id (ObjectId), tableNumber (Number, required, unique), capacity (Number, required, min: 1), isAvailable (Boolean, default: true)	Referenced in Reservations.
Reservations	_id (ObjectId), user (ObjectId, ref: 'User', required), table (ObjectId, ref: 'Table', required), date (String,	Belongs to a User and a Table.

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Collection	Fields & Types	Relationships & Rules
	required), time (String, required), guests (Number, required), status (String, enum: ['Pending', 'Confirmed', 'Cancelled'], default: 'Pending'), createdAt, updatedAt	
Orders	_id (ObjectId), user (ObjectId, ref: 'User', required), items (Array of { menuItem: ObjectId, name: String, price: Number, quantity: Number }), totalAmount (Number, required, min: 0), status (String, enum: ['Pending', 'Preparing', 'Ready', 'Served', 'Cancelled'], default: 'Pending'), createdAt, updatedAt	Belongs to a User. Total calculated strictly on server.

API Endpoints Table (New Section 8)

Method	Endpoint	Access	Request Body	Response
POST	/api/auth/login	Public	email, password	{ token, user }
POST	/api/auth/register	Public	name, email, password	{ token, user }
GET	/api/menu	Public	None (Optional queries: category, search)	List of menu items
GET	/api/menu/:id	Public	None	Single menu item detail

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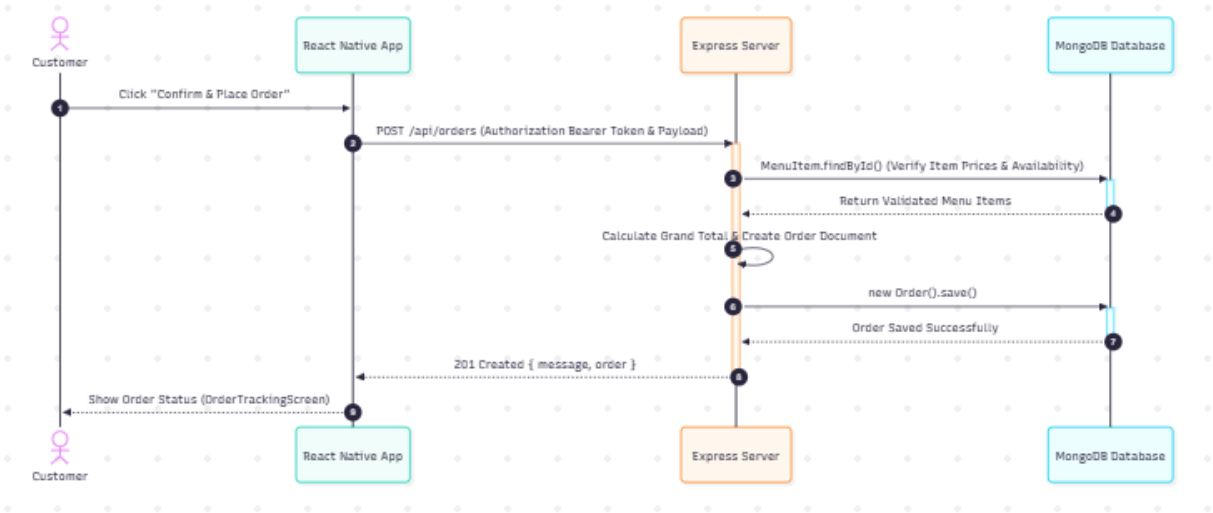
Method	Endpoint	Access	Request Body	Response
POST	/api/menu	Manager Only	name, description, price, category, image	Created menu item (201)
PUT	/api/menu/:id	Manager Only	Updated item fields	Updated menu item (200)
DELETE	/api/menu/:id	Manager Only	None	Success message (200)
POST	/api/orders	Customer	items: [{ menuItem, quantity }]	Created order object (201)
GET	/api/orders/my	Customer	None	List of current user's orders
GET	/api/orders	Manager Only	None	List of all system orders
PATCH	/api/orders/:id/status	Manager Only	status	Updated order object
GET	/api/tables	Public	None	List of dining tables
POST	/api/reservations	Customer	tableId, date, time, guests	Created reservation (201)

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Method	Endpoint	Access	Request Body	Response
GET	/api/reservations/my	Customer	None	List of user's reservations
GET	/api/reservations	Manager Only	None	List of all reservations
PATCH	/api/reservations/:id	Manager Only	status	Updated reservation
GET	/api/health	Public	None	{ status: "ok", time: ... }

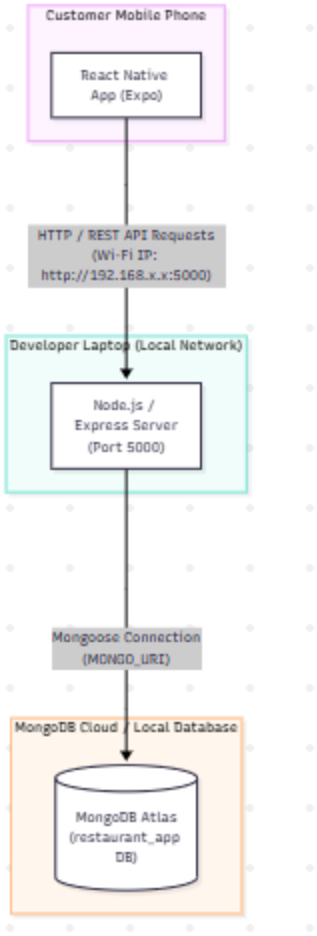
Diagrams:

Sequence Diagram:



Deployment Diagram:

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